

13.7 Logarithm. General Power. Principal Value

We finally introduce the *complex logarithm*, which is more complicated than the real logarithm (which it includes as a special case) and historically puzzled mathematicians for some time (so if you first get puzzled—which need not happen!—be patient and work through this section with extra care).

The **natural logarithm** of $z = x + iy$ is denoted by $\ln z$ (sometimes also by $\log z$) and is defined as the inverse of the exponential function; that is, $w = \ln z$ is defined for $z \neq 0$ by the relation

$$e^w = z.$$

(Note that $z = 0$ is impossible, since $e^w \neq 0$ for all w ; see Sec. 13.5.) If we set $w = u + iv$ and $z = re^{i\theta}$, this becomes

$$e^w = e^{u+iv} = re^{i\theta}.$$

Now, from Sec. 13.5, we know that e^{u+iv} has the absolute value e^u and the argument v . These must be equal to the absolute value and argument on the right:

$$e^u = r, \quad v = \theta.$$

$e^u = r$ gives $u = \ln r$, where $\ln r$ is the familiar *real* natural logarithm of the positive number $r = |z|$. Hence $w = u + iv = \ln z$ is given by

$$(1) \quad \ln z = \ln r + i\theta \quad (r = |z| > 0, \quad \theta = \arg z).$$

Now comes an important point (without analog in real calculus). Since the argument of z is determined only up to integer multiples of 2π , *the complex natural logarithm $\ln z$ ($z \neq 0$) is infinitely many-valued.*

The value of $\ln z$ corresponding to the principal value $\text{Arg } z$ (see Sec. 13.2) is denoted by $\text{Ln } z$ (Ln with capital L) and is called the **principal value** of $\ln z$. Thus

$$(2) \quad \text{Ln } z = \ln |z| + i \text{Arg } z \quad (z \neq 0).$$

The uniqueness of $\text{Arg } z$ for given z ($\neq 0$) implies that $\text{Ln } z$ is single-valued, that is, a function in the usual sense. Since the other values of $\arg z$ differ by integer multiples of 2π , the other values of $\ln z$ are given by

$$(3) \quad \ln z = \text{Ln } z \pm 2n\pi i \quad (n = 1, 2, \dots).$$

They all have the same real part, and their imaginary parts differ by integer multiples of 2π .

If z is positive real, then $\text{Arg } z = 0$, and $\text{Ln } z$ becomes identical with the real natural logarithm known from calculus. If z is negative real (so that the natural logarithm of calculus is not defined!), then $\text{Arg } z = \pi$ and

$$\text{Ln } z = \ln |z| + \pi i \quad (z \text{ negative real}).$$

From (1) and $e^{\ln r} = r$ for positive real r we obtain

$$(4a) \quad e^{\ln z} = z$$

as expected, but since $\arg(e^z) = y \pm 2n\pi$ is multivalued, so is

$$(4b) \quad \ln(e^z) = z \pm 2n\pi i, \quad n = 0, 1, \dots$$

EXAMPLE Natural Logarithm. Principal Value

$\ln 1 = 0, \pm 2\pi i, \pm 4\pi i, \dots$	$\text{Ln } 1 = 0$
$\ln 4 = 1.386294 \pm 2n\pi i$	$\text{Ln } 4 = 1.386294$
$\ln(-1) = \pm \pi i, \pm 3\pi i, \pm 5\pi i, \dots$	$\text{Ln}(-1) = \pi i$
$\ln(-4) = 1.386294 \pm (2n+1)\pi i$	$\text{Ln}(-4) = 1.386294 + \pi i$
$\ln i = \pi i/2, -3\pi i/2, 5\pi i/2, \dots$	$\text{Ln } i = \pi i/2$
$\ln 4i = 1.386294 + \pi i/2 \pm 2n\pi i$	$\text{Ln } 4i = 1.386294 + \pi i/2$
$\ln(-4i) = 1.386294 - \pi i/2 \pm 2n\pi i$	$\text{Ln}(-4i) = 1.386294 - \pi i/2$
$\ln(3-4i) = \ln 5 + i \arg(3-4i)$ $= 1.609438 - 0.927295i \pm 2n\pi i$	$\text{Ln}(3-4i) = 1.609438 - 0.927295i$ (Fig. 337)

Eq (4b) $\ln(e^z) ??$

Sol:-

As we know that

$$\ln(z) = \ln r + i\theta \rightarrow \textcircled{A}$$

where

$$r = |z|, \quad \theta = \arg z$$

$$\text{so } |e^z| = |e^{x+iy}| = |e^x(\cos y + i \sin y)| = |e^x \cos y + i e^x \sin y| = \sqrt{e^{2x} \cos^2 y + e^{2x} \sin^2 y} = \sqrt{e^{2x} (1)} = e^x \quad \text{As } \cos^2 y + \sin^2 y = 1$$

$$\text{so } r = |e^z| = e^x$$

$$\text{and } \arg \text{ of } e^z = \arg \text{ of } [e^x(\cos y + i \sin y)] = \tan^{-1} \left[\frac{e^x \sin y}{e^x \cos y} \right] = \tan^{-1} \tan y = y$$

$$\text{so } \theta = y \pm 2n\pi$$

$$\text{from } \textcircled{A} \quad \ln(e^z) = \ln(e^x) + i(y \pm 2n\pi)$$

$$= 2 \ln e + iy \pm 2n\pi i \quad \because \ln e = 1$$

$$\ln(e^z) = (x + iy) \pm 2n\pi i$$

$$\boxed{\ln(e^z) = z \pm 2n\pi i} \rightarrow (4b) \text{ where } n = 0, 1, 2, \dots$$

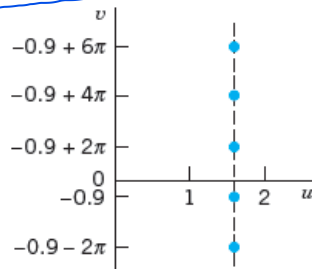


Fig. 337. Some values of $\ln(3 - 4i)$ in Example 1

The familiar relations for the natural logarithm continue to hold for complex values, that is,

$$(5) \quad (a) \quad \ln(z_1 z_2) = \ln z_1 + \ln z_2, \quad (b) \quad \ln(z_1/z_2) = \ln z_1 - \ln z_2$$

but these relations are to be understood in the sense that each value of one side is also contained among the values of the other side; see the next example.

EXAMPLE 2 Illustration of the Functional Relation (5) in Complex

Let

$$z_1 = z_2 = e^{\pi i} = -1.$$

If we take the principal values

$$\operatorname{Ln} z_1 = \operatorname{Ln} z_2 = \pi i,$$

then (5a) holds provided we write $\ln(z_1 z_2) = \ln 1 = 2\pi i$; however, it is not true for the principal value, $\operatorname{Ln}(z_1 z_2) = \operatorname{Ln} 1 = 0$. ■

THEOREM 1

Analyticity of the Logarithm

For every $n = 0, \pm 1, \pm 2, \dots$ formula (3) defines a function, which is analytic, except at 0 and on the negative real axis, and has the derivative

$$(6) \quad (\ln z)' = \frac{1}{z} \quad (z \text{ not } 0 \text{ or negative real}).$$

PROOF We show that the Cauchy–Riemann equations are satisfied. From (1)–(3) we have

$$\ln z = \ln r + i(\theta + c) = \frac{1}{2} \ln (x^2 + y^2) + i \left(\arctan \frac{y}{x} + c \right)$$

where the constant c is a multiple of 2π . By differentiation,

$$u_x = \frac{x}{x^2 + y^2} = v_y = \frac{1}{1 + (y/x)^2} \cdot \frac{1}{x}$$

$$u_y = \frac{y}{x^2 + y^2} = -v_x = -\frac{1}{1 + (y/x)^2} \left(-\frac{y}{x^2} \right).$$

Hence the Cauchy–Riemann equations hold. [Confirm this by using these equations in polar form, which we did not use since we proved them only in the problems (to Sec. 13.4).] Formula (4) in Sec. 13.4 now gives (6),

$$(\ln z)' = u_x + iv_x = \frac{x}{x^2 + y^2} + i \frac{1}{1 + (y/x)^2} \left(-\frac{y}{x^2} \right) = \frac{x - iy}{x^2 + y^2} = \frac{1}{z}. \quad \blacksquare$$

Each of the infinitely many functions in (3) is called a **branch** of the logarithm. The negative real axis is known as a **branch cut** and is usually graphed as shown in Fig. 338. The branch for $n = 0$ is called the **principal branch** of $\ln z$.

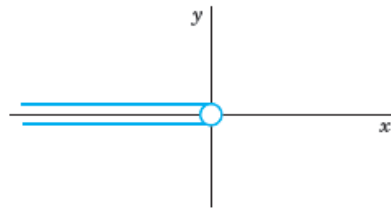


Fig. 338. Branch cut for $\ln z$

General Powers

General powers of a complex number $z = x + iy$ are defined by the formula

$$(7) \quad z^c = e^{c \ln z} \quad (c \text{ complex, } z \neq 0).$$

Since $\ln z$ is infinitely many-valued, z^c will, in general, be multivalued. The particular value

$$z^c = e^{c \operatorname{Ln} z}$$

is called the **principal value** of z^c .

If $c = n = 1, 2, \dots$, then z^n is single-valued and identical with the usual n th power of z .
If $c = -1, -2, \dots$, the situation is similar.

If $c = 1/n$, where $n = 2, 3, \dots$, then

$$z^c = \sqrt[n]{z} = e^{(1/n) \ln z} \quad (z \neq 0),$$

the exponent is determined up to multiples of $2\pi i/n$ and we obtain the n distinct values of the n th root, in agreement with the result in Sec. 13.2. If $c = p/q$, the quotient of two positive integers, the situation is similar, and z^c has only finitely many distinct values. However, if c is real irrational or genuinely complex, then z^c is infinitely many-valued.

EXAMPLE 3 General Power

$$i^i = e^{i \ln i} = \exp(i \ln i) = \exp\left[i\left(\frac{\pi}{2}i \pm 2n\pi i\right)\right] = e^{-(\pi/2) \mp 2n\pi}.$$

All these values are real, and the principal value ($n = 0$) is $e^{-\pi/2}$.

Similarly, by direct calculation and multiplying out in the exponent,

$$\begin{aligned} (1+i)^{2-i} &= \exp[(2-i) \ln(1+i)] = \exp[(2-i) \{\ln \sqrt{2} + \frac{1}{4}\pi i \pm 2n\pi i\}] \\ &= 2e^{\pi/4 \pm 2n\pi} [\sin(\frac{1}{2} \ln 2) + i \cos(\frac{1}{2} \ln 2)]. \end{aligned}$$

Recommended Exercise; 13.7 (Advanced Engineering Mathematics by
Elwin Kreyszig)

It is a *convention* that for real positive $z = x$ the expression z^c means $e^{c \ln x}$ where $\ln x$ is the elementary real natural logarithm (that is, the principal value $\text{Ln } z$ ($z = x > 0$) in the sense of our definition). Also, if $z = e$, the base of the natural logarithm, $z^c = e^c$ is *conventionally* regarded as the unique value obtained from (1) in Sec. 13.5.

From (7) we see that for any complex number a ,

$$(8) \quad a^z = e^{z \ln a}.$$

We have now introduced the complex functions needed in practical work, some of them (e^z , $\cos z$, $\sin z$, $\cosh z$, $\sinh z$) entire (Sec. 13.5), some of them ($\tan z$, $\cot z$, $\tanh z$, $\coth z$) analytic except at certain points, and one of them ($\ln z$) splitting up into infinitely many functions, each analytic except at 0 and on the negative real axis.

For the **inverse trigonometric** and **hyperbolic functions** see the problem set.

Basic definitions

A **simple curve** can be drawn without lifting the pencil from the paper, and without passing through any point twice.

A **closed curve** has its starting and ending points the same, and is also drawn without lifting the pencil from the paper.



Simple;
closed



Simple; not
closed



Not simple;
closed



Not simple;
not closed

Definition. A region D is **connected** if any two points in D can be joined by a path in D .



Connected



Connected



not connected

Definition. A region D is **simply connected** if it is connected and for every simple closed curve C in D , C encloses only points of D . Informally, simply connected means connected with no holes.



Simply connected.



→ not simply connected b/c not connected.



Connected
But not simply
connected



→ connected

But not simply connected.

14.1 Line Integral in the Complex Plane

As in calculus, in complex analysis we distinguish between definite integrals and indefinite integrals or antiderivatives. Here an **indefinite integral** is a function whose derivative equals a given analytic function in a region. By inverting known differentiation formulas we may find many types of indefinite integrals.

Complex definite integrals are called (complex) **line integrals**. They are written

$$\int_C f(z) dz.$$

Here the **integrand** $f(z)$ is integrated over a given curve C or a portion of it (an *arc*, but we shall say “*curve*” in either case, for simplicity). This curve C in the complex plane is called the **path of integration**. We may represent C by a parametric representation

$$(1) \quad z(t) = x(t) + iy(t) \quad (a \leq t \leq b).$$

The sense of increasing t is called the **positive sense** on C , and we say that C is **oriented** by (1).

For instance, $z(t) = t + 3it$ ($0 \leq t \leq 2$) gives a portion (a segment) of the line $y = 3x$. The function $z(t) = 4 \cos t + 4i \sin t$ ($-\pi \leq t \leq \pi$) represents the circle $|z| = 4$, and so on. More examples follow below.

We assume C to be a **smooth curve**, that is, C has a continuous and nonzero derivative

$$\dot{z}(t) = \frac{dz}{dt} = \dot{x}(t) + i\dot{y}(t)$$

at each point. Geometrically this means that C has everywhere a continuously turning tangent, as follows directly from the definition

$$\dot{z}(t) = \lim_{\Delta t \rightarrow 0} \frac{z(t + \Delta t) - z(t)}{\Delta t} \quad (\text{Fig. 339}).$$

Here we use a dot since a prime ' denotes the derivative with respect to z .

Definition of the Complex Line Integral

This is similar to the method in calculus. Let C be a smooth curve in the complex plane given by (1), and let $f(z)$ be a continuous function given (at least) at each point of C . We now subdivide (we “*partition*”) the interval $a \leq t \leq b$ in (1) by points

$$t_0 (= a), \quad t_1, \quad \dots, \quad t_{n-1}, \quad t_n (= b)$$

where $t_0 < t_1 < \dots < t_n$. To this subdivision there corresponds a subdivision of C by points

$$z_0, \quad z_1, \quad \dots, \quad z_{n-1}, \quad z_n (= Z) \quad (\text{Fig. 340}),$$

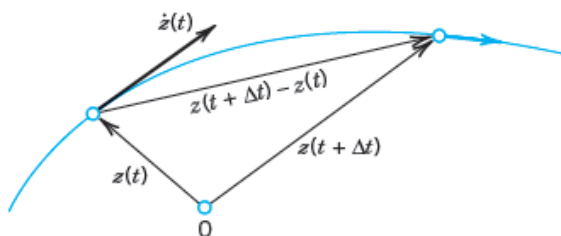


Fig. 339. Tangent vector $\dot{z}(t)$ of a curve C in the complex plane given by $z(t)$. The arrowhead on the curve indicates the *positive sense* (sense of increasing t)

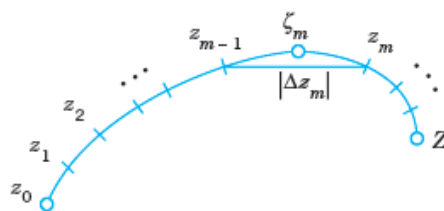


Fig. 340. Complex line integral

where $z_j = z(t_j)$. On each portion of subdivision of C we choose an arbitrary point, say, a point ζ_1 between z_0 and z_1 (that is, $\zeta_1 = z(t)$ where t satisfies $t_0 \leq t \leq t_1$), a point ζ_2 between z_1 and z_2 , etc. Then we form the sum

$$(2) \quad S_n = \sum_{m=1}^n f(\zeta_m) \Delta z_m \quad \text{where} \quad \Delta z_m = z_m - z_{m-1}.$$

We do this for each $n = 2, 3, \dots$ in a completely independent manner, but so that the greatest $|\Delta t_m| = |t_m - t_{m-1}|$ approaches zero as $n \rightarrow \infty$. This implies that the greatest $|\Delta z_m|$ also approaches zero. Indeed, it cannot exceed the length of the arc of C from z_{m-1} to z_m and the latter goes to zero since the arc length of the smooth curve C is a continuous function of t . The limit of the sequence of complex numbers $S_2, S_3, \dots$ thus obtained is called the **line integral** (or simply the *integral*) of $f(z)$ over the path of integration C with the orientation given by (1). This line integral is denoted by

$$(3) \quad \int_C f(z) dz, \quad \text{or by} \quad \oint_C f(z) dz$$

if C is a **closed path** (one whose terminal point Z coincides with its initial point z_0 , as for a circle or for a curve shaped like an 8).

General Assumption. *All paths of integration for complex line integrals are assumed to be piecewise smooth, that is, they consist of finitely many smooth curves joined end to end.*

Basic Properties Directly Implied by the Definition

1. **Linearity.** Integration is a **linear operation**, that is, we can integrate sums term by term and can take out constant factors from under the integral sign. This means that if the integrals of f_1 and f_2 over a path C exist, so does the integral of $k_1 f_1 + k_2 f_2$ over the same path and

$$(4) \quad \int_C [k_1 f_1(z) + k_2 f_2(z)] dz = k_1 \int_C f_1(z) dz + k_2 \int_C f_2(z) dz.$$

2. **Sense reversal** in integrating over the *same* path, from z_0 to Z (left) and from Z to z_0 (right), introduces a minus sign as shown,

$$(5) \quad \int_{z_0}^Z f(z) dz = - \int_Z^{z_0} f(z) dz.$$

3. **Partitioning of path** (see Fig. 341)

$$(6) \quad \int_C f(z) dz = \int_{C_1} f(z) dz + \int_{C_2} f(z) dz.$$

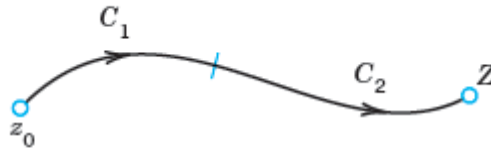


Fig. 341. Partitioning of path [formula (6)]

Existence of the Complex Line Integral

Our assumptions that $f(z)$ is continuous and C is piecewise smooth imply the existence of the line integral (3). This can be seen as follows.

As in the preceding chapter let us write $f(z) = u(x, y) + iv(x, y)$. We also set

$$\zeta_m = \xi_m + i\eta_m \quad \text{and} \quad \Delta z_m = \Delta x_m + i\Delta y_m.$$

Then (2) may be written

$$(7) \quad S_n = \sum (u + iv)(\Delta x_m + i\Delta y_m)$$

where $u = u(\zeta_m, \eta_m)$, $v = v(\zeta_m, \eta_m)$ and we sum over m from 1 to n . Performing the multiplication, we may now split up S_n into four sums:

$$S_n = \sum u \Delta x_m - \sum v \Delta y_m + i \left[\sum u \Delta y_m + \sum v \Delta x_m \right].$$

These sums are real. Since f is continuous, u and v are continuous. Hence, if we let n approach infinity in the aforementioned way, then the greatest Δx_m and Δy_m will approach zero and each sum on the right becomes a real line integral:

$$(8) \quad \begin{aligned} \lim_{n \rightarrow \infty} S_n &= \int_C f(z) dz \\ &= \int_C u dx - \int_C v dy + i \left[\int_C u dy + \int_C v dx \right]. \end{aligned}$$

This shows that under our assumptions on f and C the line integral (3) exists and its value is independent of the choice of subdivisions and intermediate points ζ_m . ■

First Evaluation Method: Indefinite Integration and Substitution of Limits

This method is the analog of the evaluation of definite integrals in calculus by the well-known formula

$$\int_a^b f(x) dx = F(b) - F(a)$$

where $[F'(x) = f(x)]$.

It is simpler than the next method, but it is suitable for analytic functions only. To formulate it, we need the following concept of general interest.

A domain D is called **simply connected** if every **simple closed curve** (closed curve without self-intersections) encloses only points of D .

For instance, a circular disk is simply connected, whereas an annulus (Sec. 13.3) is not simply connected. (Explain!)

THEOREM 1

Indefinite Integration of Analytic Functions

Let $f(z)$ be analytic in a simply connected domain D . Then there exists an indefinite integral of $f(z)$ in the domain D , that is, an analytic function $F(z)$ such that $F'(z) = f(z)$ in D , and for all paths in D joining two points z_0 and z_1 in D we have

$$(9) \quad \int_{z_0}^{z_1} f(z) dz = F(z_1) - F(z_0) \quad [F'(z) = f(z)].$$

(Note that we can write z_0 and z_1 instead of C , since we get the same value for all those C from z_0 to z_1 .)

This theorem will be proved in the next section.

Simple connectedness is quite essential in Theorem 1, as we shall see in Example 5.

Since analytic functions are our main concern, and since differentiation formulas will often help in finding $F(z)$ for a given $f(z) = F'(z)$, the present method is of great practical interest.

If $f(z)$ is entire (Sec. 13.5), we can take for D the complex plane (which is certainly simply connected).

EXAMPLE 1

$$\int_0^{1+i} z^2 dz = \frac{1}{3} z^3 \Big|_0^{1+i} = \frac{1}{3} (1+i)^3 = -\frac{2}{3} + \frac{2}{3} i$$

EXAMPLE 2

$$\int_{-\pi i}^{\pi i} \cos z dz = \sin z \Big|_{-\pi i}^{\pi i} = 2 \sin \pi i = 2i \sinh \pi = 23.097i$$

EXAMPLE 3

$$\int_{8+\pi i}^{8-3\pi i} e^{z/2} dz = 2e^{z/2} \Big|_{8+\pi i}^{8-3\pi i} = 2(e^{4-3\pi i/2} - e^{4+\pi i/2}) = 0$$

since e^z is periodic with period $2\pi i$.

EXAMPLE 4

$\int_{-i}^i \frac{dz}{z} = \text{Ln } i - \text{Ln } (-i) = \frac{i\pi}{2} - \left(-\frac{i\pi}{2}\right) = i\pi$. Here D is the complex plane without 0 and the negative real axis (where $\text{Ln } z$ is not analytic). Obviously, D is a simply connected domain.

Second Evaluation Method: Use of a Representation of a Path

This method is not restricted to analytic functions but applies to any continuous complex function.

THEOREM 2

Integration by the Use of the Path

Let C be a piecewise smooth path, represented by $z = z(t)$, where $a \leq t \leq b$. Let $f(z)$ be a continuous function on C . Then

$$(10) \quad \int_C f(z) dz = \int_a^b f[z(t)] \dot{z}(t) dt \quad \left(\dot{z} = \frac{dz}{dt} \right).$$

PROOF The left side of (10) is given by (8) in terms of real line integrals, and we show that the right side of (10) also equals (8). We have $z = x + iy$, hence $\dot{z} = \dot{x} + i\dot{y}$. We simply write u for $u[x(t), y(t)]$ and v for $v[x(t), y(t)]$. We also have $dx = \dot{x} dt$ and $dy = \dot{y} dt$. Consequently, in (10)

$$\begin{aligned} \int_a^b f[z(t)] \dot{z}(t) dt &= \int_a^b (u + iv)(\dot{x} + i\dot{y}) dt \\ &= \int_C [u dx - v dy + i(u dy + v dx)] \\ &= \int_C (u dx - v dy) + i \int_C (u dy + v dx). \end{aligned} \quad \blacksquare$$

COMMENT. In (7) and (8) of the existence proof of the complex line integral we referred to real line integrals. If one wants to avoid this, one can take (10) as a *definition* of the complex line integral.

Steps in Applying Theorem 2

- (A) Represent the path C in the form $z(t)$ ($a \leq t \leq b$).
- (B) Calculate the derivative $\dot{z}(t) = dz/dt$.
- (C) Substitute $z(t)$ for every z in $f(z)$ (hence $x(t)$ for x and $y(t)$ for y).
- (D) Integrate $f[z(t)]\dot{z}(t)$ over t from a to b .

EXAMPLE 5 A Basic Result: Integral of $1/z$ Around the Unit Circle

We show that by integrating $1/z$ counterclockwise around the unit circle (the circle of radius 1 and center 0; see Sec. 13.3) we obtain

$$(11) \quad \oint_C \frac{dz}{z} = 2\pi i \quad (C \text{ the unit circle, counterclockwise}).$$

This is a very important result that we shall need quite often.

Solution. (A) We may represent the unit circle C in Fig. 330 of Sec. 13.3 by

$$z(t) = \cos t + i \sin t = e^{it} \quad (0 \leq t \leq 2\pi),$$

so that counterclockwise integration corresponds to an increase of t from 0 to 2π .

(B) Differentiation gives $\dot{z}(t) = ie^{it}$ (chain rule!).

(C) By substitution, $f(z(t)) = 1/z(t) = e^{-it}$.

(D) From (10) we thus obtain the result

$$\oint_C \frac{dz}{z} = \int_0^{2\pi} e^{-it} ie^{it} dt = i \int_0^{2\pi} dt = 2\pi i.$$

Check this result by using $z(t) = \cos t + i \sin t$.

Simple connectedness is essential in Theorem 1. Equation (9) in Theorem 1 gives 0 for any closed path because then $z_1 = z_0$, so that $F(z_1) - F(z_0) = 0$. Now $1/z$ is not analytic at $z = 0$. But any *simply connected* domain containing the unit circle must contain $z = 0$, so that Theorem 1 does not apply—it is not enough that $1/z$ is analytic in an annulus, say, $\frac{1}{2} < |z| < \frac{3}{2}$, because an annulus is not simply connected! ■

EXAMPLE 6 Integral of $1/z^m$ with Integer Power m

Let $f(z) = (z - z_0)^m$ where m is the integer and z_0 a constant. Integrate counterclockwise around the circle C of radius ρ with center at z_0 (Fig. 342).

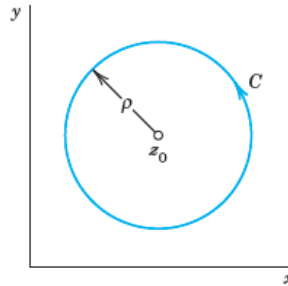


Fig. 342. Path in Example 6

Solution. We may represent C in the form

$$z(t) = z_0 + \rho(\cos t + i \sin t) = z_0 + \rho e^{it} \quad (0 \leq t \leq 2\pi).$$

Then we have

$$(z - z_0)^m = \rho^m e^{imt}, \quad dz = i\rho e^{it} dt$$

and obtain

$$\oint_C (z - z_0)^m dz = \int_0^{2\pi} \rho^m e^{imt} i\rho e^{it} dt = i\rho^{m+1} \int_0^{2\pi} e^{i(m+1)t} dt.$$

By the Euler formula (5) in Sec. 13.6 the right side equals

$$i\rho^{m+1} \left[\int_0^{2\pi} \cos(m+1)t dt + i \int_0^{2\pi} \sin(m+1)t dt \right].$$

If $m = -1$, we have $\rho^{m+1} = 1$, $\cos 0 = 1$, $\sin 0 = 0$. We thus obtain $2\pi i$. For integer $m \neq -1$ each of the two integrals is zero because we integrate over an interval of length 2π , equal to a period of sine and cosine. Hence the result is

$$(12) \quad \oint_C (z - z_0)^m dz = \begin{cases} 2\pi i & (m = -1), \\ 0 & (m \neq -1 \text{ and integer}). \end{cases}$$

Dependence on path. Now comes a very important fact. If we integrate a given function $f(z)$ from a point z_0 to a point z_1 along different paths, the integrals will in general have different values. In other words, *a complex line integral depends not only on the endpoints of the path but in general also on the path itself*. The next example gives a first impression of this, and a systematic discussion follows in the next section.

EXAMPLE 7 Integral of a Nonanalytic Function. Dependence on Path

Integrate $f(z) = \operatorname{Re} z = x$ from 0 to $1 + 2i$ (a) along C^* in Fig. 343, (b) along C consisting of C_1 and C_2 .

Solution. (a) C^* can be represented by $z(t) = t + 2it$ ($0 \leq t \leq 1$). Hence $\dot{z}(t) = 1 + 2i$ and $f[z(t)] = x(t) = t$ on C^* . We now calculate

$$\int_{C^*} \operatorname{Re} z dz = \int_0^1 t(1 + 2i) dt = \frac{1}{2}(1 + 2i) = \frac{1}{2} + i.$$

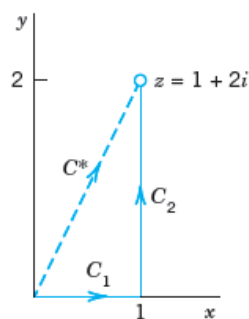


Fig. 343. Paths in Example 7

(b) We now have

$$C_1: z(t) = t, \quad \dot{z}(t) = 1, \quad f(z(t)) = x(t) = t \quad (0 \leq t \leq 1)$$

$$C_2: z(t) = 1 + it, \quad \dot{z}(t) = i, \quad f(z(t)) = x(t) = 1 \quad (0 \leq t \leq 2).$$

Using (6) we calculate

$$\int_C \operatorname{Re} z \, dz = \int_{C_1} \operatorname{Re} z \, dz + \int_{C_2} \operatorname{Re} z \, dz = \int_0^1 t \, dt + \int_0^2 1 \cdot i \, dt = \frac{1}{2} + 2i.$$

Note that this result differs from the result in (a). ■

